

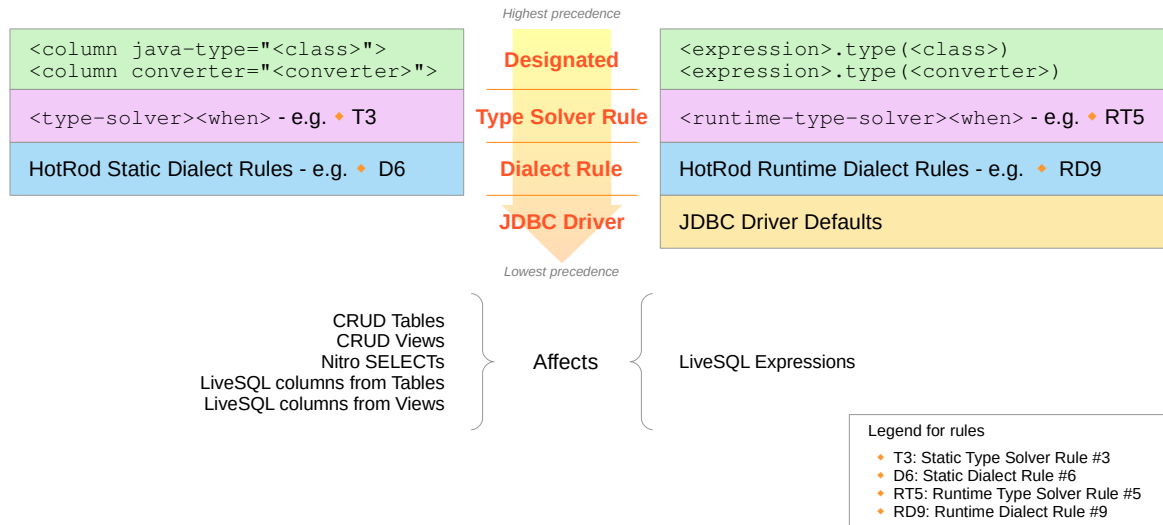
HotRod Type Resolution

STATIC TYPE SOLVER

types decided during the layer generation

RUNTIME TYPE SOLVER

types decided at runtime



Example

layer.xml:

```
<type-solver>
  <when test="typeName == 'TIMESTAMP'" java-type="java.sql.Timestamp" />
  <when test="typeName == 'TINYINT'" java-type="Double" />
  <when test="typeName == 'CHARACTER VARYING ARRAY' and name.endsWith('_TAX_CODES')"
    converter="TaxCodesArrayConverter" />
</type-solver>

<runtime-type-solver>
  <when test="columnName == 'dom'" java-type="Long" />
</runtime-type-solver>

<table name="invoice">
  <column name="created" java-type="java.sql.Timestamp" />
</table>
```

App.java:

```
sql.select(
  i.id, // STATIC_DIALECT_RULE
  i.created, // STATIC_DESIGNATED
  i.invoiceTaxCodes, // STATIC_TYPESOLVER_RULE (converter)
  i.amount, // STATIC_DIALECT_RULE
  i.paid, // STATIC_DIALECT_RULE

  i.amount.mult(1.30).as("ag1").type(Double.class), // RUNTIME_DESIGNATED
  sql.caseWhen(i.amount.ge(500), "Y").elseValue("N").end().as("vip").type(this.ynBooleanConverter),
  // RUNTIME_DESIGNATED (converter)
  i.created.extract(DateTimeField.DAY).as("dom"), // RUNTIME_TYPESOLVER_RULE
  sql.caseWhen(i.paid.eq("Y"), i.amount).elseValue(0).end().as("paidAmount"), // RUNTIME_DIALECT_RULE
  h2.randomUUID().as("globalId") // RUNTIME_JDBC_DRIVER_DEFAULT
).from(...)
```

Logging:

```
SELECT
-- the following columns come from a table, so they always use STATIC type resolution
a.id as "id",
a.created as "created",
a.invoice_tax_codes as "invoiceTaxCodes",
a.amount as "amount",
a.paid as "paid",
-- the following columns are LiveSQL expressions, so they use RUNTIME type resolution
a.amount * 1.30 as "ag1",
case when (a.amount >= 200) then 'Y' else 'N' end as "vip",
extract(day from a.created) as "dom",
case when (a.paid = 'Y') then a.amount else 0 end as "paidAmount",
RANDOM_UUID() as "globalId"
FROM invoice a
WHERE a.id = ?
--- Query Columns (10) ---
* 1 id: java.lang.Integer, source: STATIC_DIALECT_RULE, rule #D9
* 2 created: java.sql.Timestamp, source: STATIC_DESIGNATED
* 3 invoiceTaxCodes: String[] (converter TaxCodesArrayConverter), source: STATIC_TYPESOLVER_RULE, rule #T3
* 4 amount: java.math.BigDecimal, source: STATIC_DIALECT_RULE, rule #D1
* 5 paid: java.lang.String, source: STATIC_DIALECT_RULE, rule #D13
* 6 ag1: java.lang.Double, source: RUNTIME_DESIGNATED
* 7 vip: Boolean (converter YNBooleanConverter), source: RUNTIME_DESIGNATED
* 8 dom: java.lang.Long, source: RUNTIME_TYPESOLVER_RULE, rule #RT1
* 9 paidAmount: java.math.BigDecimal, source: RUNTIME_DIALECT_RULE, rule #RD1
* 10 globalId: java.util.UUID, source: RUNTIME_JDBC_DRIVER_DEFAULT
```